A Integral feedback model

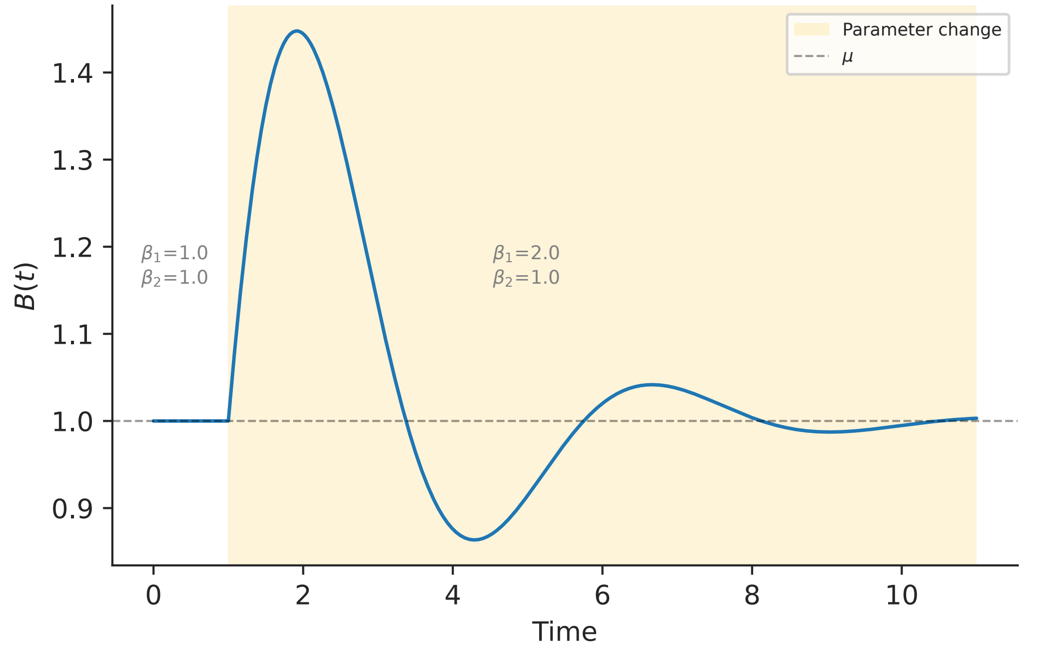
$$\frac{dA}{dt} = \alpha\,(\mu - B)$$

$$\frac{dB}{dt} = \beta_1 A - \beta_2 B$$

$$B_{ss} = \mu$$

Steady-state of B depends only on μ, and is thus robust to β_1 and β_2.

B Integral feedback ensures B converges to μ after parameter change

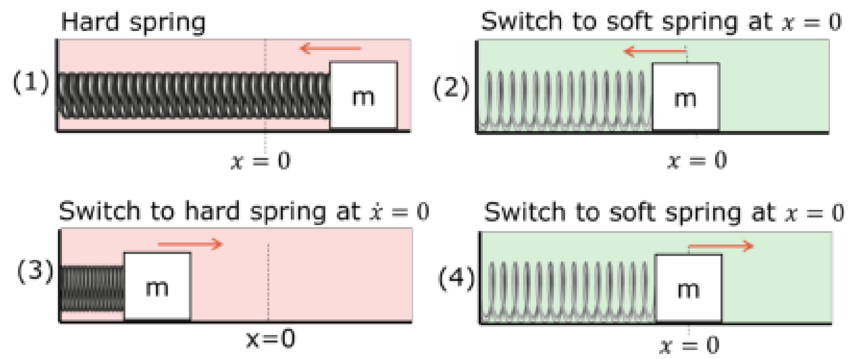


C Integral feedback model is equivalent to a mass on a spring

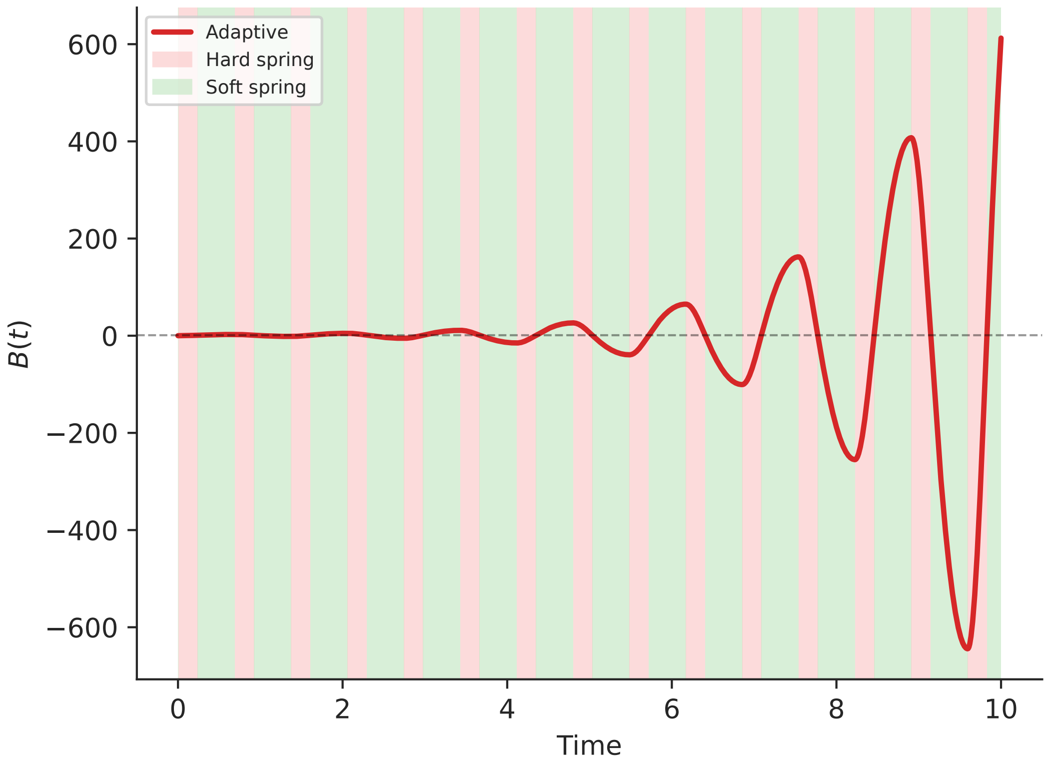
$$\frac{d^2B}{dt^2} + \beta_2\frac{dB}{dt} + \alpha\beta_1 B = \alpha\beta_1\mu$$

mass $m = 1$
damping $c = \beta_2$
spring constant $k = \alpha\beta_1$

D Adaptive strategy that amplifies the motion of a mass on a spring



E Adaptive strategy drives unbounded growth of B in AB model



F Calcium model

$P = k_0(R_0 - Ca)$, $\frac{dCa}{dt} = k_1 P + k_2 D$, $\frac{dD}{dt} = k_3 P$

$Ca_{ss} = R_0$

reduces to: $\ddot{Ca} + k_1 k_0\,\dot{Ca} + k_2 k_3 k_0\,Ca = k_2 k_3 k_0\,R_0$

spring constant $k = k_2 k_3 k_0$

G Adaptive strategy drives unbounded growth of Calcium